

Artem Semidetnov

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Education

MSc	Geneva University , Mathematics	2025 – 2027
BSc	Saint-Petersburg State University , Mathematics	2021 – 2025
BSc	Neapolis University Of Pafos , Applied Computer Science JetBrains Program	2022 – 2024

Technical Skills

- **Formal verification:** Arend, dependent type theory, theorem formalization, typechecking, and IDE integration.
- **Programming languages:** Kotlin, Python, SQL, Java, Lean, Arend, Haskell.
- **AI and machine learning:** Coding-agent frameworks (LangGraph, Koog), model fine-tuning, code completion, and MCP protocol.
- **Spoken languages:** Russian (native), English C1 (IELTS 8.0/9), French (conversational).

Work Experience

JetBrains , Research Intern in HoTT and Dependent Types Lab	Remote
• Built AI-assisted development features for Arend, including MCP servers exposing typechecking and IDE functionality; contributed to the Arend plugin for IntelliJ IDEA. • Built an experimental LLM coding agent for generating Arend code using the Koog framework. • Fine-tuned models for Arend code completion.	Nov 2025 – Apr 2026
JetBrains , Research Intern in HoTT and Dependent Types Lab	Pafos, Cyprus
• Contributed formalized results in algebra and homotopy type theory to the official Arend library. • Formalized the Eckmann-Hilton argument, Eilenberg-MacLane spaces, and homology of types. • Formalized group automorphisms, Schur’s Lemma, Maschke’s Lemma, and characterizations of group actions; selected contributions shipped in Arend 1.10.	June 2024 – Sept 2024
IPONWEB (acquired by Criteo) , Machine Learning Intern	Pafos, Cyprus
• Reverse-engineered the behavior of a Criteo ML system that analyzes websites and produces word2vec representations; evaluated potential applications.	May 2023 – Sept 2023

AI-Assisted Mathematical Research

On Some More Problems from the Kourouka Notebook	Aug 2026
Vasily Ionin, Artem Semidetnov https://doi.org/10.48550/arXiv.2608.29219	
The stable commutator length of a relator is not a one-relator group invariant	Aug 2026
Artem Semidetnov https://doi.org/10.48550/arXiv.2608.21465	

Publications

The operad associated to a crossed simplicial group	Dec 2025
Artem Semidetnov https://doi.org/10.48550/arXiv.2512.14498	
On the geometry of free nilpotent groups	May 2021
Artem Semidetnov , Ruslan Magdiev https://doi.org/10.48550/arXiv.2106.00095	